

ECON 166: International Trade

Section 1

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Trade vs. Autarky, with Ricardian assumptions

Suppose consumer preferences in both “Home” and “Foreign” can be described by $U(X, Y) = XY$.

In “Home” there are **four** units of labor, each of which can produce either **three** units of X or **six** units of Y .

In “Foreign” there are **eight** units of labor, each of which can produce either **one** unit of X or **four** units of Y .

(Labor and goods are infinitely divisible, so that decimal (not just integer) answers are possible.)

Home Economy

In “Home” there are **four** units of labor, each of which can produce either **three** units of X or **six** units of Y .

$$\bar{L} = 4$$

$$MPL_X = 3$$

$$MPL_Y = 6$$

Home Economy

In “Home” there are **four** units of labor, each of which can produce either **three** units of X or **six** units of Y .

$$\bar{L} = 4$$

$$MPL_X = 3$$

$$MPL_Y = 6$$

$$\implies X = 3L_X$$

$$\implies Y = 6L_Y$$

$$L_X + L_Y = \bar{L}$$

Foreign Economy

In “Foreign” there are **eight** units of labor, each of which can produce either **one** unit of X or **four** units of Y .

$$\bar{L}^* = 8$$

$$MPL_X^* = 1$$

$$MPL_Y^* = 4$$

Foreign Economy

In “Foreign” there are **eight** units of labor, each of which can produce either **one** unit of X or **four** units of Y .

$$\bar{L}^* = 8$$

$$MPL_X^* = 1$$

$$MPL_Y^* = 4$$

$$\implies X^* = L_X^*$$

$$\implies Y^* = 4L_Y^*$$

$$L_X^* + L_Y^* = \bar{L}^*$$

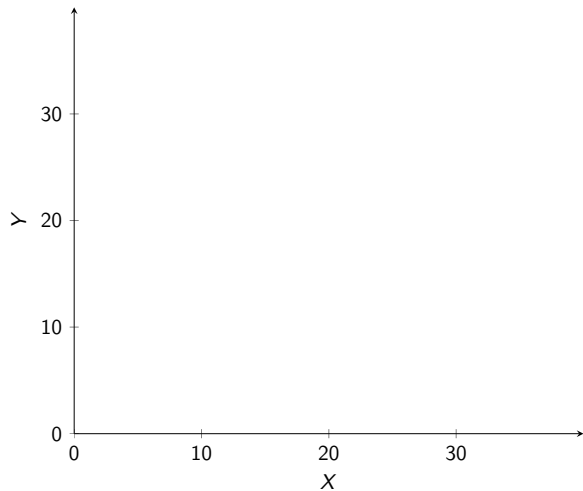
Solve for the autarky competitive equilibrium

In each country, determine:

1. the quantity of X and Y consumed and produced
2. the utility level attained by consumers
3. the price ratio P_X/P_Y

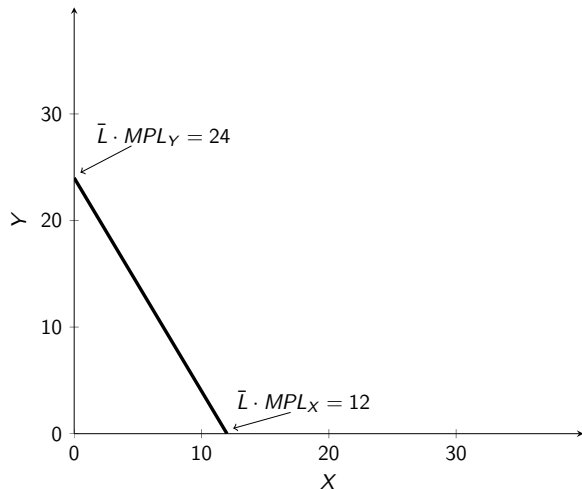
Solve for the autarky competitive equilibrium: Home

Draw the PPF



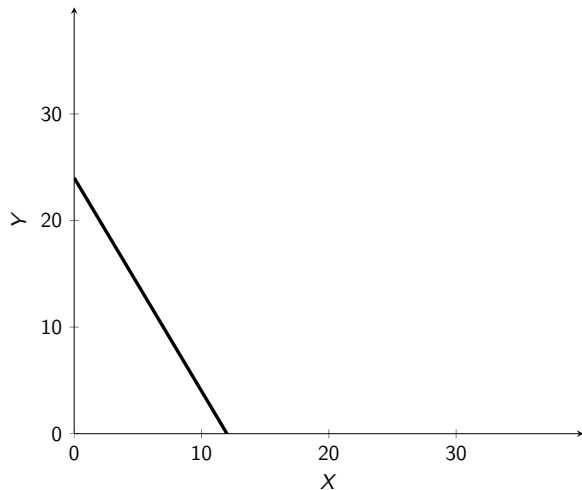
Solve for the autarky competitive equilibrium: Home

Draw the PPF



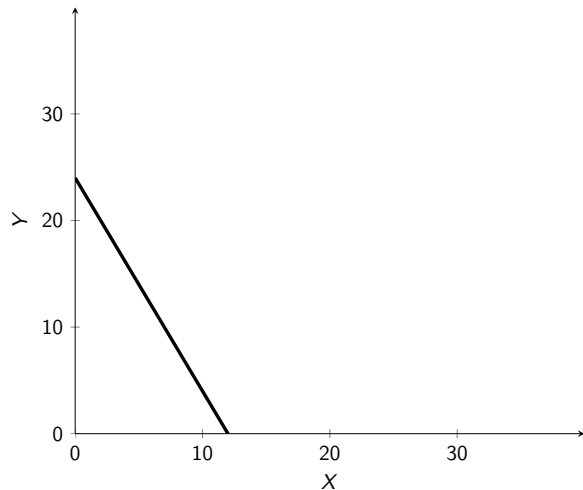
Solve for the autarky competitive equilibrium: Home

Solve the Consumer's Problem



Solve for the autarky competitive equilibrium: Home

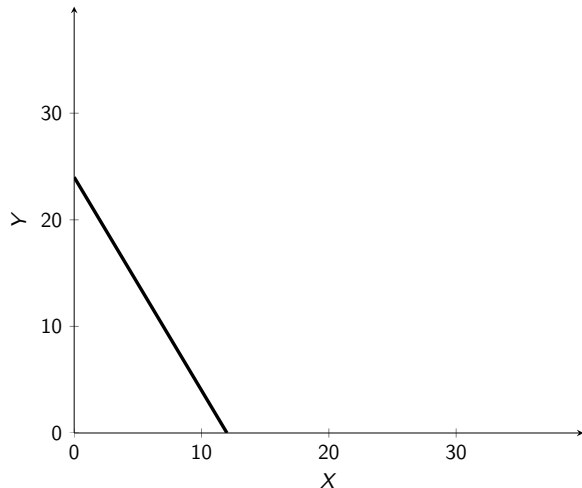
Solve the Consumer's Problem



$$\begin{aligned} \max_{X,Y} \quad & XY \\ \text{s.t.} \quad & \frac{1}{3}X + \frac{1}{6}Y = \bar{L} \end{aligned}$$

Solve for the autarky competitive equilibrium: Home

Solve the Consumer's Problem



$$\max_{X,Y} XY$$

$$\text{s.t. } \frac{1}{3}X + \frac{1}{6}Y = \bar{L}$$

Solve using Lagrangian or recall:

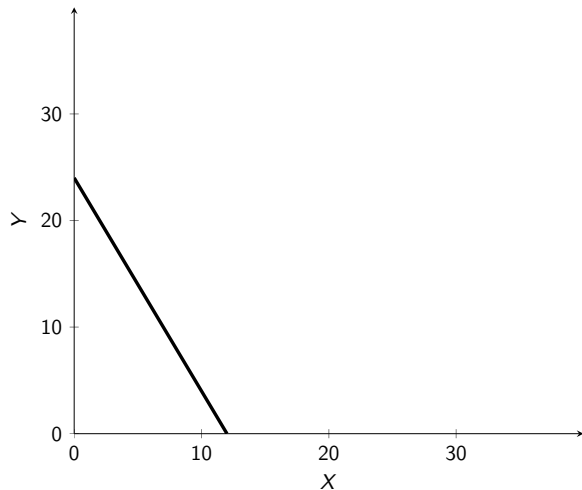
$$MRS = MRT$$

$$\frac{MU_X}{MU_Y} = \frac{MC_X}{MC_Y} \left(= \frac{MPL_Y}{MPL_X} \right)$$

$$\frac{Y}{X} = 2 \implies Y = 2X$$

Solve for the autarky competitive equilibrium: Home

Solve the Consumer's Problem



$$\max_{X,Y} XY$$

$$\text{s.t. } \frac{1}{3}X + \frac{1}{6}Y = \bar{L}$$

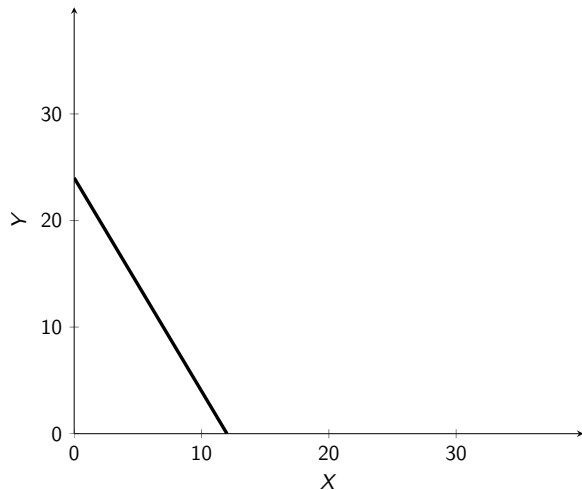
Now use the PPF:

$$\frac{1}{3}X + \frac{1}{6}Y = \bar{L}$$

$$\frac{1}{3}X + \frac{1}{6}(2X) = 4$$

Solve for the autarky competitive equilibrium: Home

Solve the Consumer's Problem



$$\max_{X,Y} XY$$

$$\text{s.t. } \frac{1}{3}X + \frac{1}{6}Y = \bar{L}$$

Now use the PPF:

$$\frac{1}{3}X + \frac{1}{6}Y = \bar{L}$$

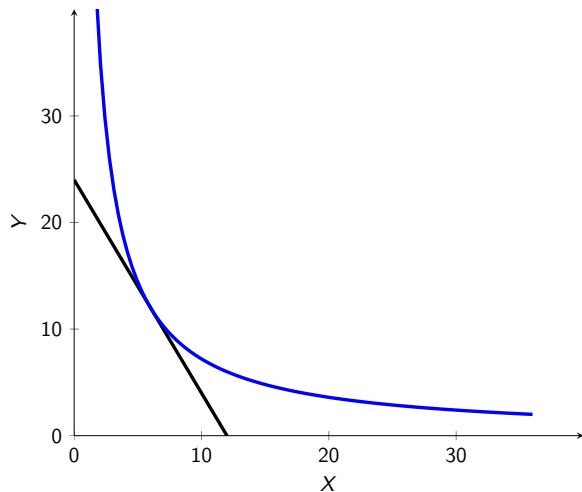
$$\frac{1}{3}X + \frac{1}{6}(2X) = 4$$

$$X = 6$$

$$Y = 12$$

Solve for the autarky competitive equilibrium: Home

Solve the Consumer's Problem



$$\begin{aligned} \max_{X,Y} \quad & XY \\ \text{s.t.} \quad & \frac{1}{3}X + \frac{1}{6}Y = \bar{L} \end{aligned}$$

Now use the PPF:

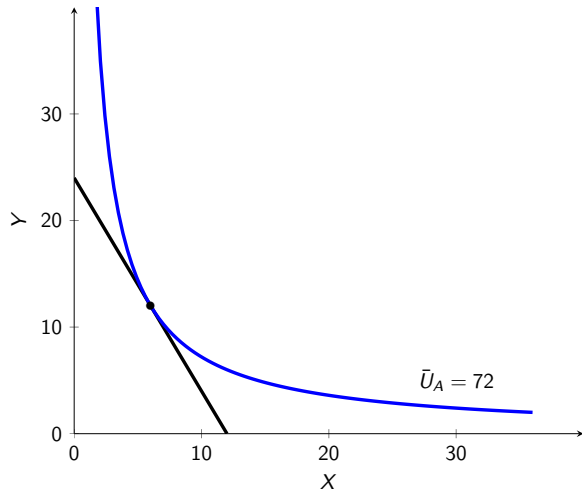
$$\begin{aligned} \frac{1}{3}X + \frac{1}{6}Y &= \bar{L} \\ \frac{1}{3}X + \frac{1}{6}(2X) &= 4 \end{aligned}$$

$$X = 6$$

$$Y = 12$$

Solve for the autarky competitive equilibrium: Home

Solve the Consumer's Problem



$$\max_{X,Y} XY$$

$$\text{s.t. } \frac{1}{3}X + \frac{1}{6}Y = \bar{L}$$

Now use the PPF:

$$\frac{1}{3}X + \frac{1}{6}Y = \bar{L}$$

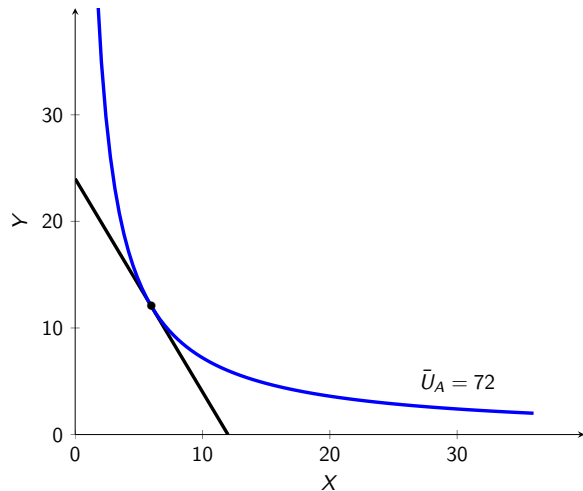
$$\frac{1}{3}X + \frac{1}{6}(2X) = 4$$

$$X = 6$$

$$Y = 12$$

Solve for the autarky competitive equilibrium: Home

Solve the Consumer's Problem



In autarky, consumption (and production) is

$$(X, Y) = (6, 12).$$

Utility is

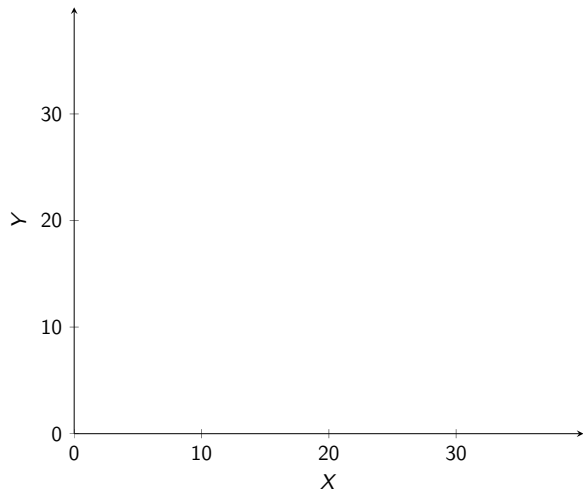
$$U(X, Y) = 72.$$

And finally, the price ratio is:

$$\frac{P_X}{P_Y} = 2$$

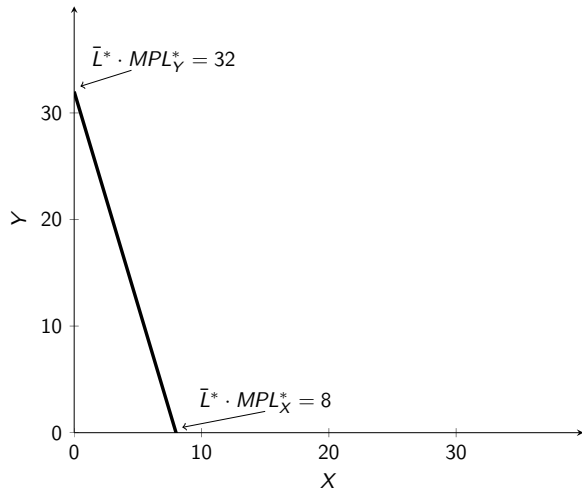
Solve for the autarky competitive equilibrium: Foreign

Draw the PPF



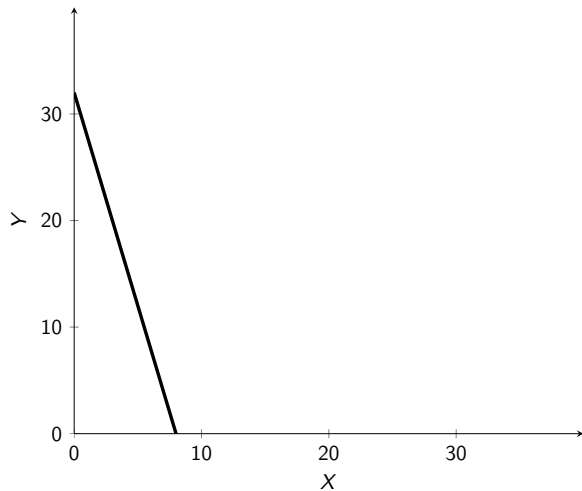
Solve for the autarky competitive equilibrium: Foreign

Draw the PPF



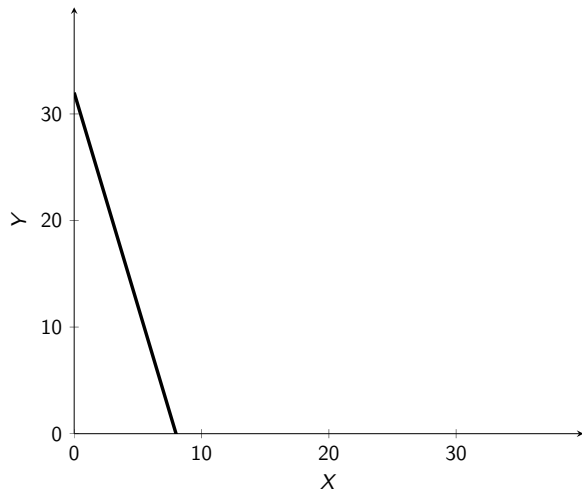
Solve for the autarky competitive equilibrium: Foreign

Solve the Consumer's Problem



Solve for the autarky competitive equilibrium: Foreign

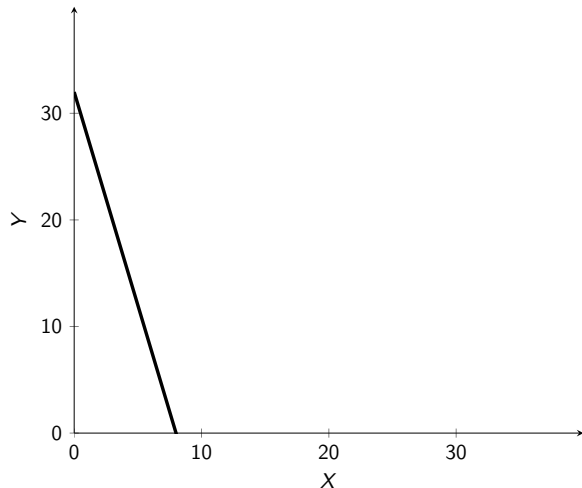
Solve the Consumer's Problem



$$\begin{aligned} \max_{X^*, Y^*} \quad & X^* Y^* \\ \text{s.t.} \quad & X^* + \frac{1}{4} Y^* = \bar{L}^* \end{aligned}$$

Solve for the autarky competitive equilibrium: Foreign

Solve the Consumer's Problem



$$\max_{X^*, Y^*} X^* Y^*$$

$$\text{s.t. } X^* + \frac{1}{4}Y^* = \bar{L}^*$$

Solve using Lagrangian or recall:

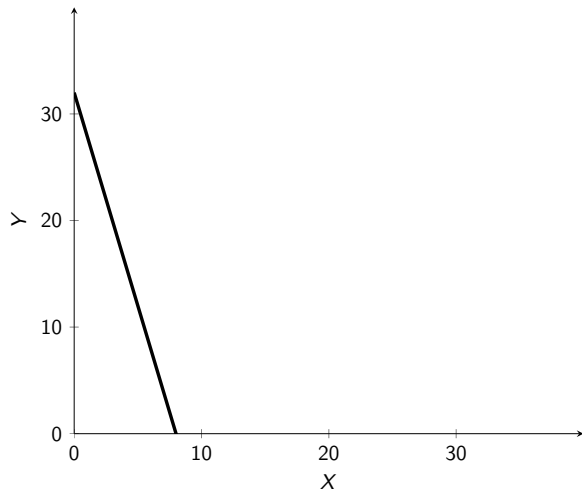
$$MRS^* = MRT^*$$

$$\frac{MU_X^*}{MU_Y^*} = \frac{MC_X^*}{MC_Y^*} \left(= \frac{MPL_Y^*}{MPL_X^*} \right)$$

$$\frac{Y^*}{X^*} = 4 \implies Y^* = 4X^*$$

Solve for the autarky competitive equilibrium: Foreign

Solve the Consumer's Problem



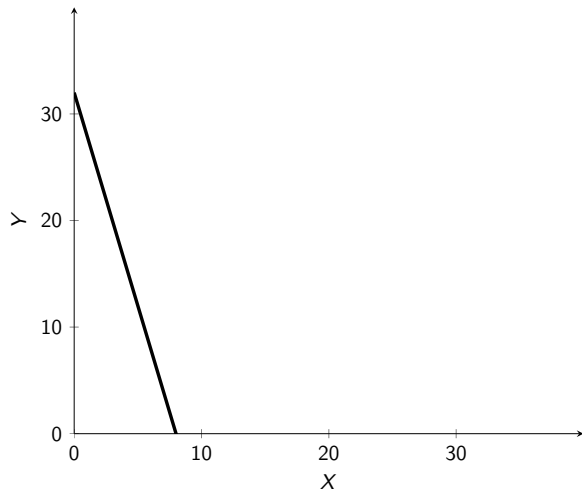
$$\begin{aligned} \max_{X^*, Y^*} \quad & X^* Y^* \\ \text{s.t.} \quad & X^* + \frac{1}{4} Y^* = \bar{L}^* \end{aligned}$$

Now use the PPF:

$$\begin{aligned} X^* + \frac{1}{4} Y^* &= \bar{L}^* \\ X^* + \frac{1}{4} (4X^*) &= 8 \end{aligned}$$

Solve for the autarky competitive equilibrium: Foreign

Solve the Consumer's Problem



$$\begin{aligned} \max_{X^*, Y^*} \quad & X^* Y^* \\ \text{s.t.} \quad & X^* + \frac{1}{4} Y^* = \bar{L}^* \end{aligned}$$

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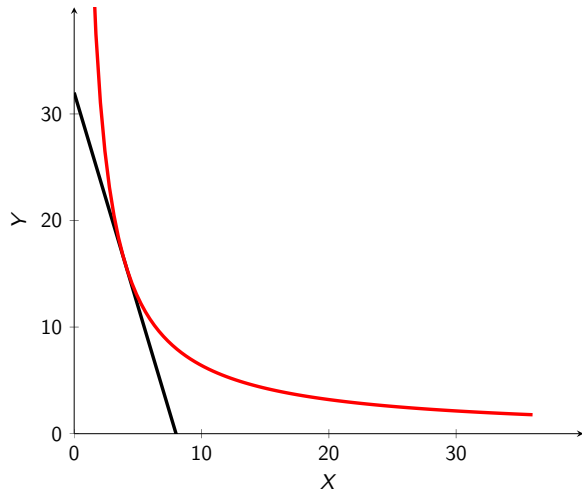
$$\begin{aligned} X^* + \frac{1}{4} Y^* &= \bar{L}^* \\ X^* + \frac{1}{4} (4X^*) &= 8 \end{aligned}$$

$$X^* = 4$$

$$Y^* = 16$$

Solve for the autarky competitive equilibrium: Foreign

Solve the Consumer's Problem



$$\begin{aligned} \max_{X^*, Y^*} \quad & X^* Y^* \\ \text{s.t.} \quad & X^* + \frac{1}{4} Y^* = \bar{L}^* \end{aligned}$$

Now use the PPF:

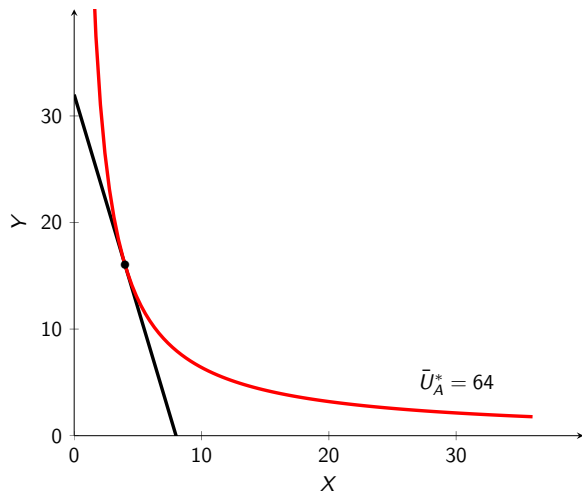
$$\begin{aligned} X^* + \frac{1}{4} Y^* &= \bar{L}^* \\ X^* + \frac{1}{4} (4X^*) &= 8 \end{aligned}$$

$$X^* = 4$$

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Solve for the autarky competitive equilibrium: Foreign

Solve the Consumer's Problem



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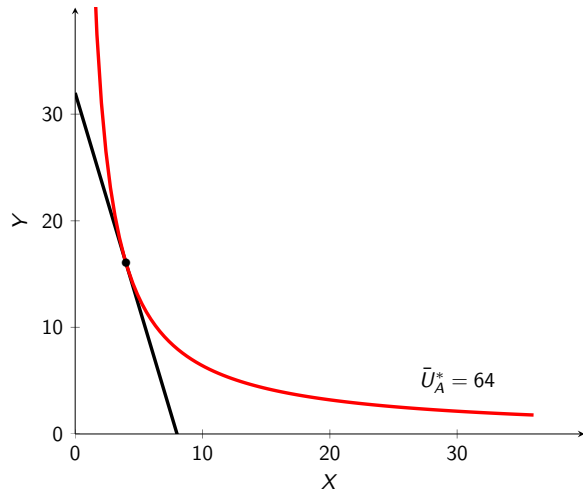
$$\begin{aligned} X^* + \frac{1}{4} Y^* &= \bar{L}^* \\ X^* + \frac{1}{4} (4X^*) &= 8 \end{aligned}$$

$$X^* = 4$$

$$Y^* = 16$$

Solve for the autarky competitive equilibrium: Foreign

Solve the Consumer's Problem



In autarky, consumption (and production) is

$$(X^*, Y^*) = (4, 16).$$

Utility is

$$U(X^*, Y^*) = 64.$$

And finally, the price ratio is:

$$\frac{P_X^*}{P_Y^*} = 4.$$

Which country has the **absolute** advantage in producing each good? The **comparative** advantage?

Absolute Advantage

Comparative Advantage

Which country has the **absolute** advantage in producing each good? The **comparative** advantage?

Absolute Advantage

Home has the absolute advantage in producing both X and Y , since

$$3 = MPL_X > MPL_X^* = 1 \text{ and } 6 = MPL_Y > MPL_Y^* = 4$$

Comparative Advantage

Which country has the **absolute** advantage in producing each good? The **comparative** advantage?

Absolute Advantage

Home has the absolute advantage in producing both X and Y , since

$$3 = MPL_X > MPL_X^* = 1 \text{ and } 6 = MPL_Y > MPL_Y^* = 4$$

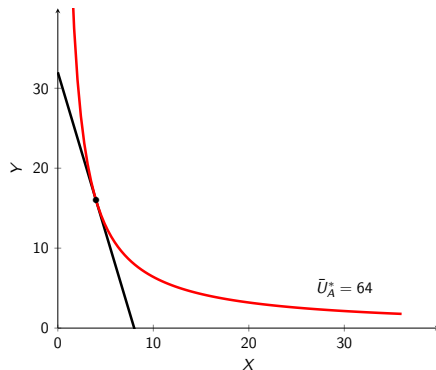
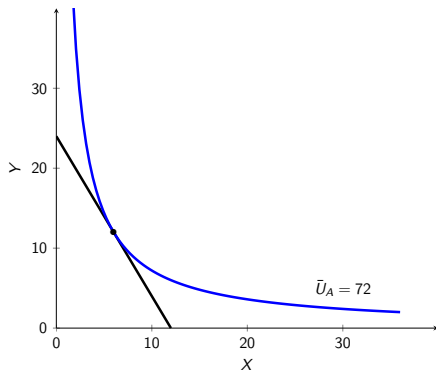
Comparative Advantage

Home has the comparative advantage in producing X , since

$$2 = \frac{P_X}{P_Y} < \frac{P_X^*}{P_Y^*} = 4$$

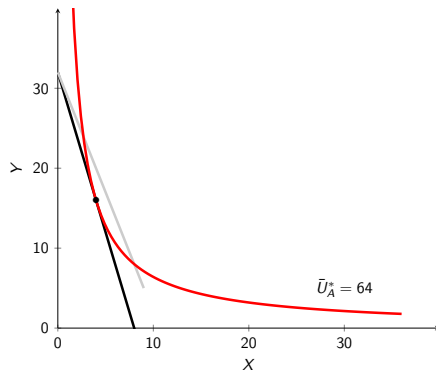
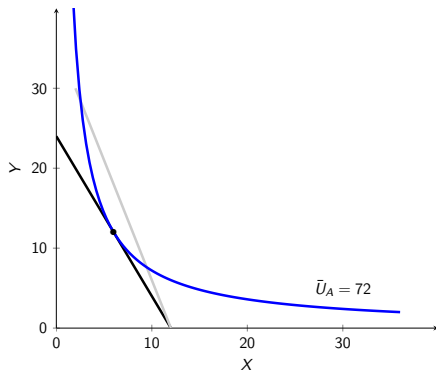
and Foreign has the comparative advantage in producing Y .

For what range of world prices P_X/P_Y could Home and Foreign engage in mutually beneficial trade?



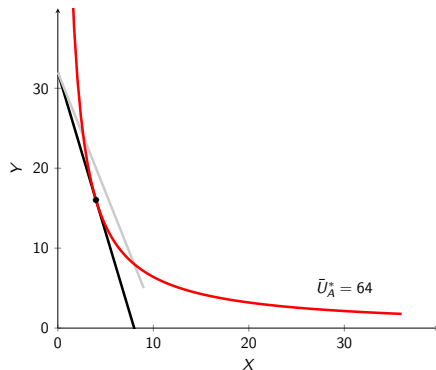
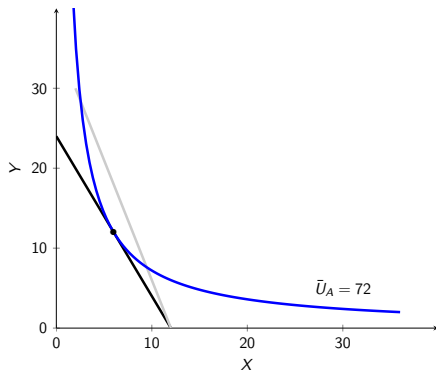
Mutually beneficial trade occurs if

For what range of world prices P_X/P_Y could Home and Foreign engage in mutually beneficial trade?



Mutually beneficial trade occurs if $2 < \left(\frac{P_X}{P_Y} \right)^W < 4$.

For what range of world prices P_X/P_Y could Home and Foreign engage in mutually beneficial trade?



Mutually beneficial trade occurs if $2 < \left(\frac{P_X}{P_Y} \right)^W < 4$.
Home will export X and Foreign will export Y .

Derive the import demand and export supply curves for X .

Export supply for X

With trade, Home solves:

$$\begin{aligned} \max_{X,Y} \quad & XY \\ \text{s.t.} \quad & P_X X + P_Y Y = 12P_X \end{aligned}$$

Derive the import demand and export supply curves for X .

Export supply for X

With trade, Home solves:

$$\begin{aligned} \max_{X,Y} \quad & XY \\ \text{s.t.} \quad & P_X X + P_Y Y = 12P_X \end{aligned}$$

Optimal consumption implies

$$MRS = \frac{Y}{X} = \frac{P_X}{P_Y} \implies P_X X = P_Y Y$$

Substitute into the budget constraint to get optimal domestic consumption of X :

$$\begin{aligned} P_X X + P_Y Y &= 12P_X \\ 2P_X X &= 12P_X \\ X &= 6 \end{aligned}$$

Derive the import demand and export supply curves for X .

Export supply for X

If $\frac{P_X}{P_Y} > 2$, then Home produces 12 units of X and consumes 6 units of X . Then

$$\begin{aligned}\text{Export supply} &= \text{domestic production} - \text{domestic consumption} \\ &= 12 - 6 \\ &= 6\end{aligned}$$

If $\frac{P_X}{P_Y} < 2$, then Home produces 6 units of X , consumes 6 units of X , and exports nothing.

Derive the import demand and export supply curves for X .

Import demand for X

With trade, Foreign solves:

$$\begin{aligned} \max_{X^*, Y^*} \quad & X^* Y^* \\ \text{s.t.} \quad & P_X X^* + P_Y Y^* = 32P_Y \end{aligned}$$

Derive the import demand and export supply curves for X .

Import demand for X

With trade, Foreign solves:

$$\begin{aligned} \max_{X^*, Y^*} \quad & X^* Y^* \\ \text{s.t.} \quad & P_X X^* + P_Y Y^* = 32P_Y \end{aligned}$$

Optimal consumption implies

$$MRS = \frac{Y^*}{X^*} = \frac{P_X}{P_Y} \implies P_X X^* = P_Y Y^*$$

Substitute into the budget constraint to get optimal domestic consumption of X^* :

$$\begin{aligned} P_X X^* + P_Y Y^* &= 32P_Y \\ 2P_X X^* &= 32P_Y^* \\ X^* &= 16 \frac{P_Y}{P_X} = \frac{16}{P_X/P_Y} \end{aligned}$$

Derive the import demand and export supply curves for X .

Import demand for X

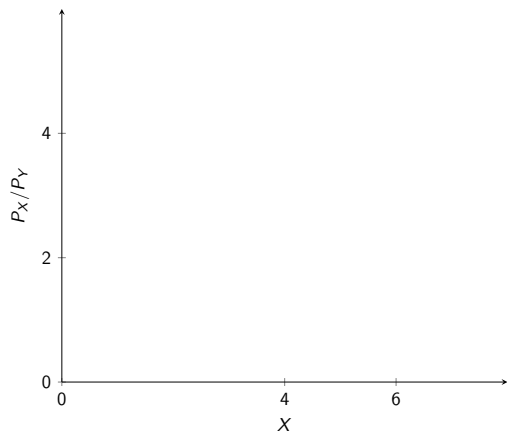
If $\frac{P_X}{P_Y} < 4$, then Foreign produces 0 units of X and consumes $\frac{16}{P_X/P_Y}$ units of X . Then

Import demand = domestic demand – domestic production

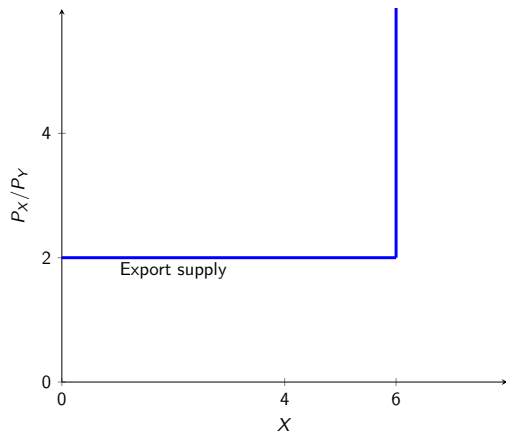
$$\begin{aligned} &= \frac{16}{P_X/P_Y} - 0 \\ &= \frac{16}{P_X/P_Y} \end{aligned}$$

If $\frac{P_X}{P_Y} > 4$, then Foreign produces 4 units of X , consumes 4 units of X , and imports nothing.

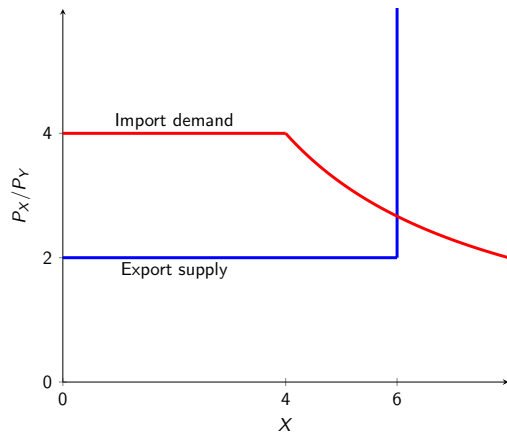
Competitive equilibrium price ratio P_X/P_Y with international trade



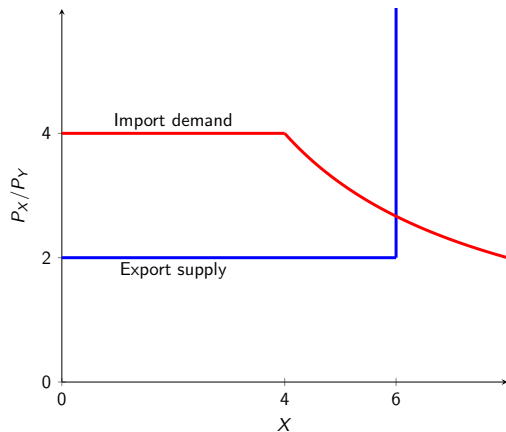
Competitive equilibrium price ratio P_X/P_Y with international trade



Competitive equilibrium price ratio P_X/P_Y with international trade



Competitive equilibrium price ratio P_X/P_Y with international trade



Solving for the price ratio:

demand = supply

$$\frac{16}{P_X/P_Y} = 6$$

$$\frac{P_X}{P_Y} = \frac{8}{3}$$

International Trade Equilibrium

How much X and Y will each country produce and consume in the international trade equilibrium?

Home

Produces 12 units of X
and consumes

$$X = 6$$

$$Y = \frac{P_X}{P_Y}X = \frac{8}{3}(6) = 16,$$

yielding utility

$$U(X, Y) = XY = 6(16) = 96.$$

Foreign

Produces 32 units of Y
and consumes

$$X^* = \frac{16}{P_X/P_Y} = \frac{16}{8/3} = 6$$

$$Y^* = \frac{P_X}{P_Y}X^* = \frac{8}{3}(6) = 16,$$

yielding utility

$$U(X^*, Y^*) = X^*Y^* = 6(16) = 96.$$

International Trade Equilibrium

How much X and Y will each country produce and consume in the international trade equilibrium?

Home

Home exports 6 units of X and imports 16 units of Y , so

$$6P_X = 16P_Y \implies \frac{P_X}{P_Y} = \frac{8}{3},$$

consistent with the terms of trade computed above.

Foreign

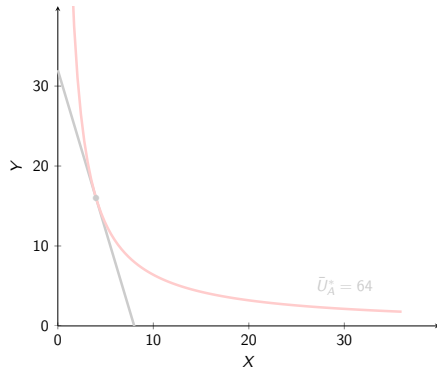
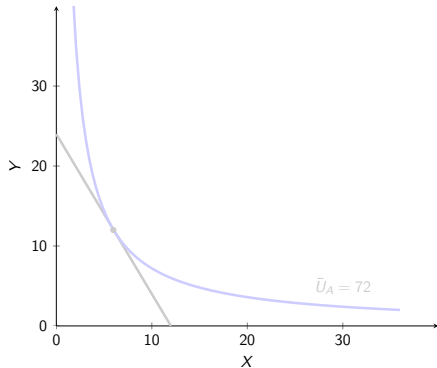
Foreign exports 16 units of Y and imports 6 units of X , so

$$6P_X = 16P_Y \implies \frac{P_X}{P_Y} = \frac{8}{3},$$

consistent with the terms of trade computed above.

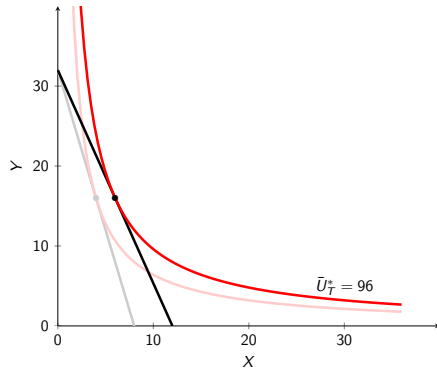
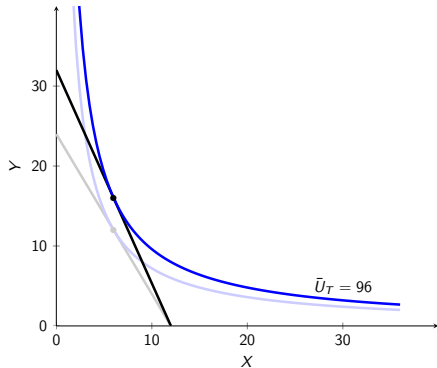
International Trade Equilibrium

How much X and Y will each country produce and consume in the international trade equilibrium?



International Trade Equilibrium

How much X and Y will each country produce and consume in the international trade equilibrium?



How does international trade affect real wages in each country?

Autarky

Real wages in Home: $MPL_X = 3$ units of X or $MPL_Y = 6$ units of Y

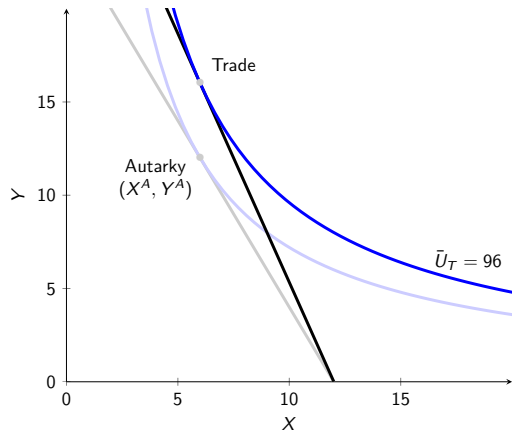
Real wages in Foreign: $MPL_Y^* = 4$ units of Y or $MPL_X^* = 1$ units of X

With Trade

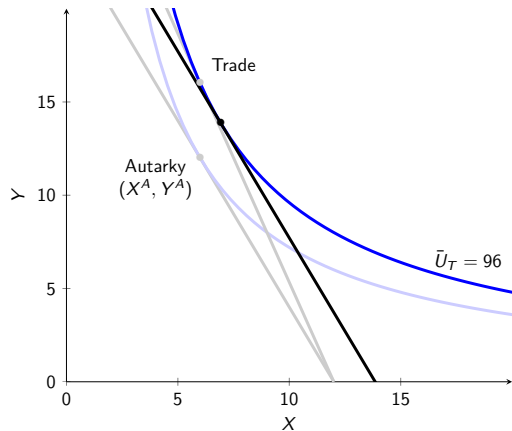
Real wages in Home: $MPL_X = 3$ units of X or $\frac{P_X}{P_Y} MPL_X = 8$ units of Y

Real wages in Foreign: $MPL_Y^* = 4$ units of Y or $\frac{P_Y}{P_X} MPL_Y^* = \frac{3}{2}$ units of X

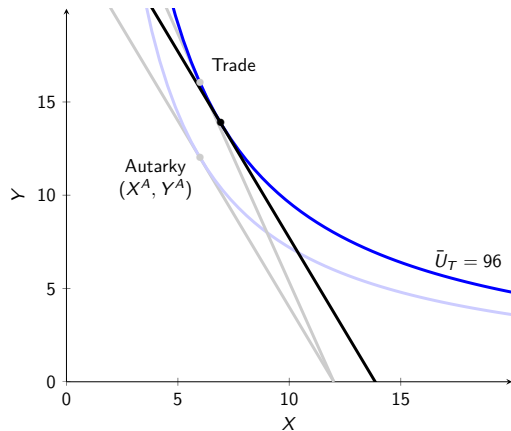
Suppose a trade war causes a complete cessation of trade between Home and Foreign.



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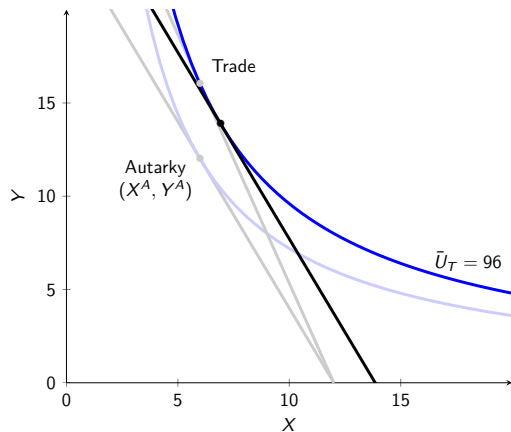


Suppose a trade war causes a complete cessation of trade between Home and Foreign.



$$\begin{aligned} \max_{X,Y} \quad & XY \\ \text{s.t.} \quad & P_X X + P_Y Y = P_X X^A + P_Y Y^A + CV \\ & XY = \bar{U}_T \end{aligned}$$

Suppose a trade war causes a complete cessation of trade between Home and Foreign.



$$\max_{X,Y} XY$$

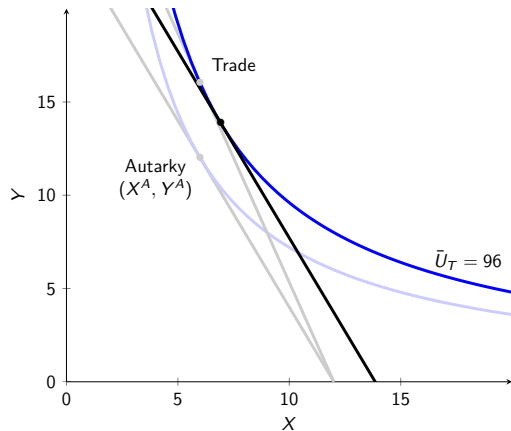
$$\text{s.t. } P_X X + P_Y Y = P_X X^A + P_Y Y^A + CV$$

$$XY = \bar{U}_T$$

Use $MRS = \frac{P_X}{P_Y}$ to see that

$$\frac{Y}{X} = 2 \implies Y = 2X.$$

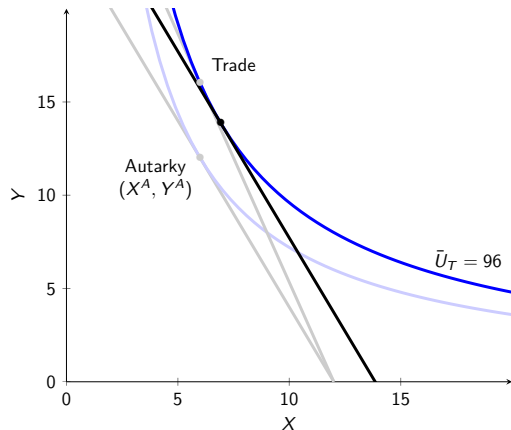
Suppose a trade war causes a complete cessation of trade between Home and Foreign.



To attain $\bar{U}_T = 96$, consumption must be

$$XY = X(2X) = 96 \implies X = 4\sqrt{3}$$
$$Y = 2X = 8\sqrt{3}$$

Suppose a trade war causes a complete cessation of trade between Home and Foreign.



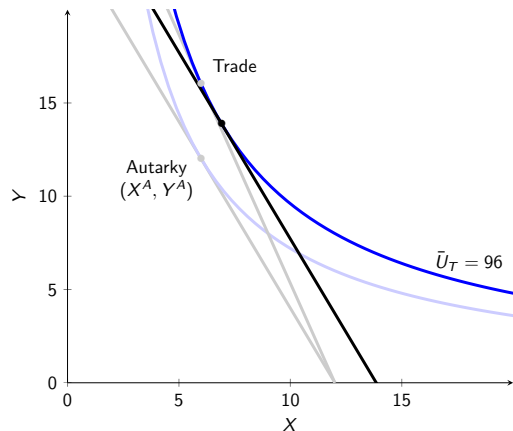
Optimal consumption is $(X, Y) = (4\sqrt{3}, 8\sqrt{3})$.

Since Home is already producing $(X^A, Y^A) = (6, 12)$, they only need

$$\begin{aligned} CV &= P_X(X - X^A) + P_Y(Y - Y^A) \\ &= 2(4\sqrt{3} - 6) + (8\sqrt{3} - 12) \\ &\approx 3.71 \end{aligned}$$

to achieve the same utility as under trade.

Suppose a trade war causes a complete cessation of trade between Home and Foreign.



Compare CV to total Home spending under autarky:

$$\frac{CV}{P_X^A X^A + P_Y^A Y^A} = \frac{3.71}{2 \cdot 6 + 1 \cdot 12} = \frac{3.71}{24} \approx 15.5\%$$

Compare CV to total Home spending under international trade:

$$\frac{CV}{P_X^A X^A + P_Y^A Y^A} = \frac{3.71}{\frac{8}{3} \cdot 6 + 1 \cdot 16} = \frac{3.71}{32} \approx 11.6\%$$